

Prolog Program for STUDENT-TEACHER-SUBCODE (AI)

Aim

To write a Prolog program that models the relationship between students, subject codes, and the teachers who teach those subjects, and to query the teacher for a given student.

Objectives

- To represent multiple related facts (teaches/2 and studies/2) in Prolog.
- To define a derived rule that combines two base relations.
- To use backtracking to list all student-teacher-subject relationships.
- To query the teacher of a specific student through the subject code.

Algorithm

- 1 Start.
- 2 Define facts teaches(Teacher, SubCode) for each teacher.
- 3 Define facts studies(Student, SubCode) for each student.
- 4 Define the rule student_teacher(Student, Teacher), linking the two relations through SubCode.
- 5 Use a fail-driven loop to display all student-subject-teacher combinations.
- 6 Query student_teacher for a specific student.
- 7 Prolog unifies the student's subject code with the matching teacher.
- 8 Display the resulting teacher.
- 9 Stop.

Source Code (Prolog)

```
% Prolog program - STUDENT-TEACHER-SUBCODE Database
teaches('Mr.Raj', 'CS101').
teaches('Mrs.Priya', 'CS102').
teaches('Mr.Kumar', 'CS103').
studies('Anitha', 'CS101').
studies('Bala', 'CS102').
studies('Chitra', 'CS101').
studies('Deepak', 'CS103').
% Rule: find the teacher of a given student
student_teacher(Student, Teacher) :-
    studies(Student, SubCode),
    teaches(Teacher, SubCode).
show_all :-
    studies(Student, SubCode),
    teaches(Teacher, SubCode),
    format("~w studies ~w, taught by ~w~n", [Student, SubCode, Teacher]),
    fail.
show_all.
:- write('Student-Teacher-SubCode Relations:'), nl,
   show_all,
   nl,
   student_teacher('Anitha', T),
   format("Query: student_teacher(Anitha, T) -> T = ~w~n", [T]).
```

Output

```
Student-Teacher-SubCode Relations:  
Anitha studies CS101, taught by Mr.Raj  
Bala studies CS102, taught by Mrs.Priya  
Chitra studies CS101, taught by Mr.Raj  
Deepak studies CS103, taught by Mr.Kumar  
Query: student_teacher(Anitha, T) -> T = Mr.Raj
```

Result

The Prolog program successfully modelled the Student-Teacher-SubCode relationships and correctly determined that Anitha is taught by Mr.Raj.